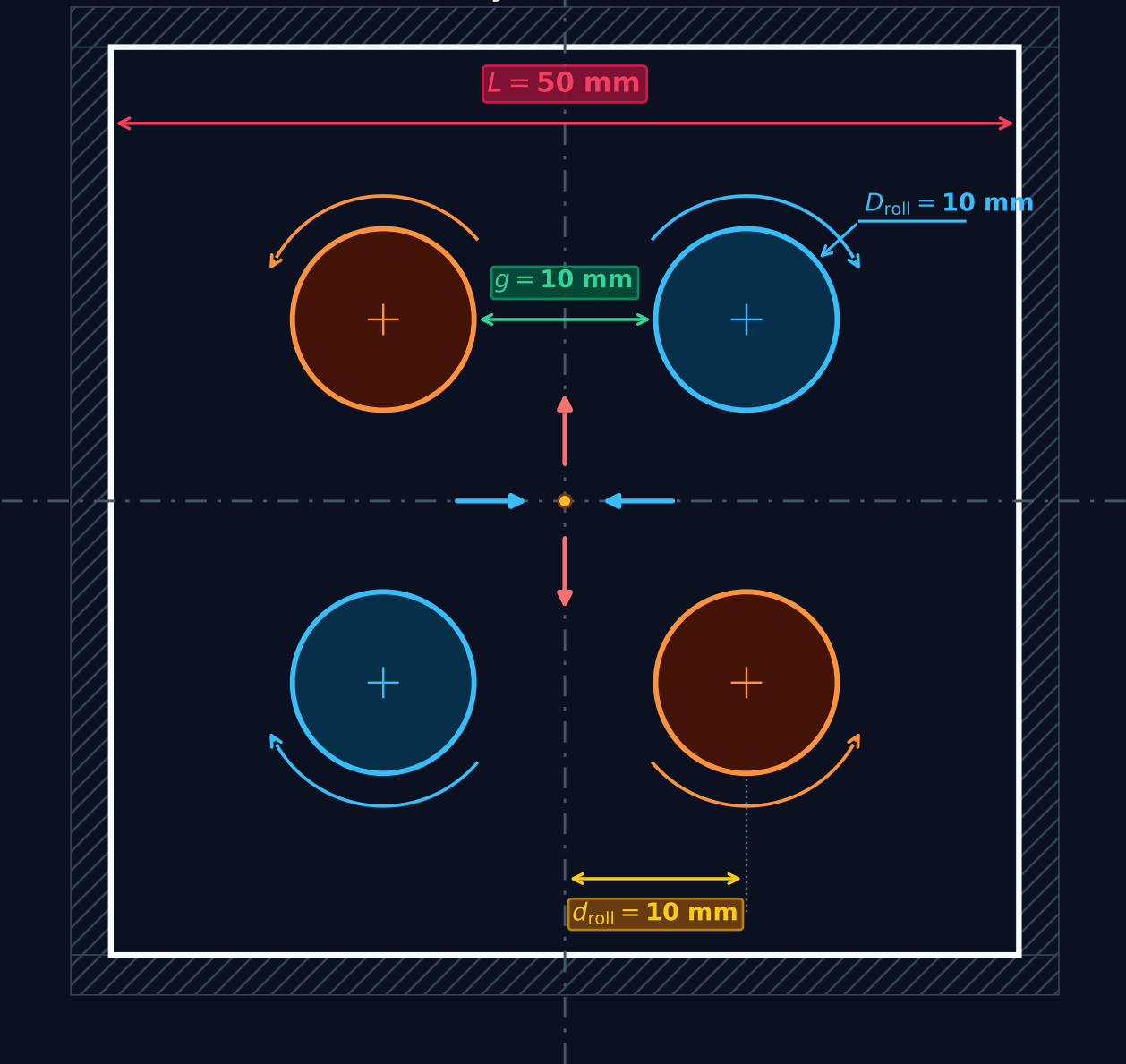


CFD Modeling: Geometric Ratios & Benchmark Setup

Canonical four-roll mill geometry compared with literature reference values

GEOMETRIC DOMAIN

Characteristic Dimensions & Symmetries



CYLINDER ASPECT RATIO (D_{roll}/L)

0.200 $[D_{\text{roll}} = 10 \text{ mm}, L = 50 \text{ mm}]$

- Lit. Value: **0.204** (D'Avino & Maffettone 2017)
- Agreement: $\Delta = -1.96\%$ (Scale consistency preserved)

GAP SPACING & POSITIONING RATIOS

Roller Positioning (d_{roll}/L) : **0.200** [Lit: 0.165]

Inter-Roll Gap $(\text{Gap}_{\text{roll}}/L)$: **0.200** [Lit: 0.126]

Roller-Wall Gap $(\text{Gap}_{\text{wall}}/L)$: **0.200** [Lit: 0.233]

• Setup Rationale: Symmetric Equidistant Design

$\text{Gap}_{\text{roll}} = \text{Gap}_{\text{wall}} = 10 \text{ mm}$ (Uniform distribution of viscous and elastic stresses)